3	
(a)	The resultant force acting on the object must be zero.
	The resultant torque about any axis is zero.
(b)	Taking moments about the hinge, $$2T\left(\frac{3}{5}\right)(0.800)=W\left(\frac{1.3}{2}\right)$$ $$2(47.5)\left(\frac{3}{5}\right)(0.800)=W\left(\frac{1.3}{2}\right)$$
	$W=70.2$ N $m=\dfrac{70.2}{9.81}=7.2$ kg (2 s.f.)

(c)	The tension in the cables has a horizontal component, while the weight has only vertical component. For the canopy to be in equilibrium, there must be a horizontal component by the hinge away from the hinge.
	The sum of the vertical component of the tensions in both cables is less than the weight, hence the force by the hinge has a vertical upward component.
4	